

Declines in the coastal ecosystems of the Great Barrier Reef (GBR) have been linked with increases in the land-based runoff of suspended sediments and nutrients and the addition of herbicides since European settlement. The [2013 Scientific Consensus Statement](#) noted that there is strong evidence that improving catchment water quality will increase the resilience of the Reef and associated ecosystems, buying some time by partially offsetting the increasing damage and stress from climate factors.

The Australian and Queensland governments established the [Reef Water Quality Protection Plan in 2003](#) to halt and reverse the decline in the quality of the water entering the GBR lagoon. Over the period 2008-09 to 2015-16, \$158 million was spent on improving agricultural management practices in Reef catchments through grants to land managers and industry. Land managers have also invested an estimated \$1.60 for each dollar provided by the Australian Government. Over \$140 million (GST exclusive) has been committed through the current Australian Government Reef Programme and a further \$40 million through the [Reef Trust](#). The Department of Agriculture has committed \$53.36 million to the Australian Government Reef Programme.

Careful targeting of expenditure is needed to achieve the desired improvements in the quality of the water flowing to the GBR lagoon. In developing the [Reef Water Quality Protection Plan 2013](#), it was agreed that actions under the Plan would include prioritising investment and knowledge. This report is the Australian Government contribution to [Reef Plan Action 3 – Investment Prioritisation](#). The report was prepared by a working group led by the Department of Agriculture and comprising of scientists from the Australian and Queensland governments, the CSIRO and Australian universities.

- [Reef Water Quality Protection Plan 2013 Prioritisation project report PDF](#)  [5.8 MB, 90 pages]
- [Reef Water Quality Protection Plan 2013 Prioritisation project report DOCX](#)  [7 MB, 90 pages]

The report provides information from the Queensland Government and James Cook University on:

- the loads of pollutants that grazing and sugar cane are estimated to contribute to the Reef,
- the likely risk to the Reef posed by these pollutants from the cane and grazing industries in particular sub-catchments, and
- the extent of practice change in cane and grazing industries; identifying the priorities for future investment which could be expected to deliver the biggest improvements in water quality.

It contains recommendations for increasing the adoption of specific practices in the cane and grazing industries, and for improvements in modelling and reporting of water quality and for further research. The results will be used by Natural Resource Management organisations to better target sub-catchments that are hotspots for pollutant delivery, and by the Department of Environment to prioritise Reef Trust funding.

Key findings:

- Land used for agriculture occupies about 82 per cent of the GBR catchment. Land used for cropping, dairy, grazing, horticulture (including bananas) and sugar cane, contributes an estimated 56, 69 and 66 per cent respectively to the estimated anthropogenic loads of the total suspended solids (TSS), particulate nitrogen (PN) and particulate phosphorus (PP), plus an

estimated 87 per cent of the anthropogenic dissolved inorganic nitrogen (DIN) and 100 per cent of the PSII herbicides delivered to the GBR lagoon.

- Grazing (75 per cent of the area of the GBR catchment) contributes an estimated 45, 43 and 45 per cent respectively to the estimated anthropogenic loads of TSS, PP and PN, and an estimated 21 per cent of the DIN load. The Burdekin and Fitzroy sub-catchments are rated as very high and high priority for investment in improving grazing management.
- Stream bank erosion, (which cannot be attributed to particular land uses but occurs largely on land used for grazing), contributes an estimated 39, 28 and 21 per cent respectively to the anthropogenic loads of TSS, PP and PN.
- Land used for sugar cane (1.3 per cent of the GBR catchment) contributes an estimated 56 and 94 per cent respectively to anthropogenic loads of DIN and PSII delivered to the GBR lagoon. The Multi-Criteria Analysis Shell for Spatial Decision Support (MCAS–S) process used for the analysis identified the Johnstone sub-catchment as very high priority. The Burdekin (mainly East Burdekin), Haughton, Herbert, Mulgrave-Russell and Tully sub-catchments are rated as high priority for investment in nutrient practices to reduce DIN loads. The Herbert, Pioneer and Plane sub-catchments are rated as very high priority for investing in improving herbicide management practices; the Haughton is rated as high and the Johnstone and O’Connell as moderate priority for investment in herbicide management.
- The impact on funding over time of changes in priorities is discussed.
- Opportunities for improving practices for better water quality outcomes are identified. In the grazing industry these include supporting adoption of better herd management practices to deliver ground cover improvements whilst improving profitability, and targeting investment to reduce sub soil loss through gullying and stream bank erosion. In the sugar cane industry, there are significant opportunities to reduce DIN loads, particularly by moving from district yield to block or zone potential yields to calculate nitrogen fertiliser applications.
- Recommendations are made on improvements in reporting, monitoring and modelling land management practices to track investment outcomes, areas identified for updating Source catchments’ modelling to reflect new understanding of sediment storage processes, and to provide spatially detailed water quality outputs, especially for the Burdekin and Fitzroy sub-catchments.
- The research needed to further improve investment targeting is identified.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/natural-resources/reef-programme>